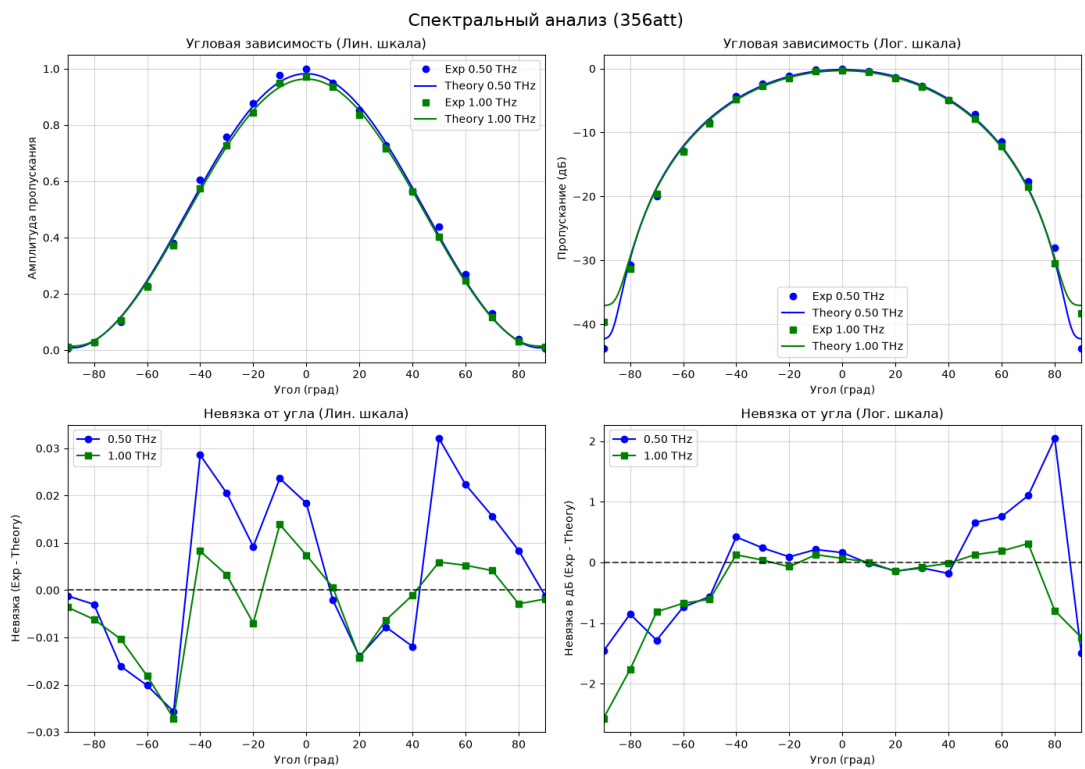


: THz-TDS

THz-Unified-Optimizer

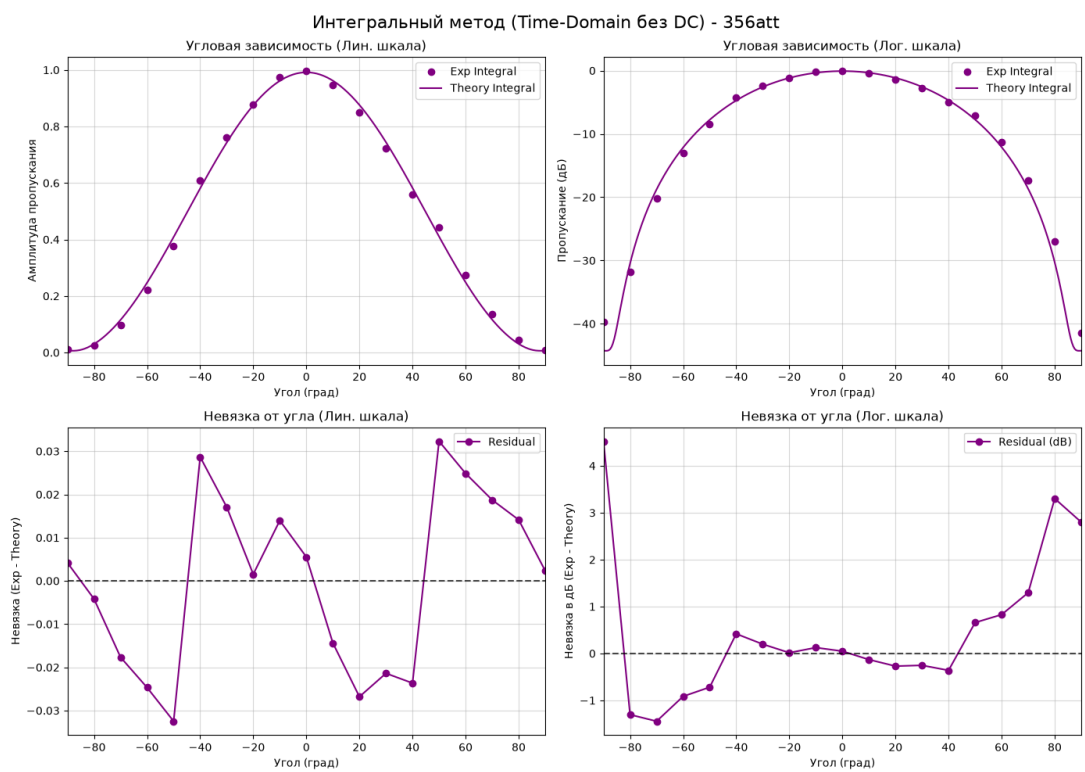
16 2026 .

1			2
1.1		(Global Average)	2
2	:		2
2.1		356att	2
2.2		series3	3
3		()	5
3.1		(RMSE)	5
3.2		(p-value)	5
3.3	2D		5
4			6
4.1			6
4.2		θ	6
5			7
5.1		($p \approx 0?$)	7
5.2		(2D-)	7
5.3	-		8
5.4			8
6			8



. 1:

356att (0.5 1.0).



. 2:

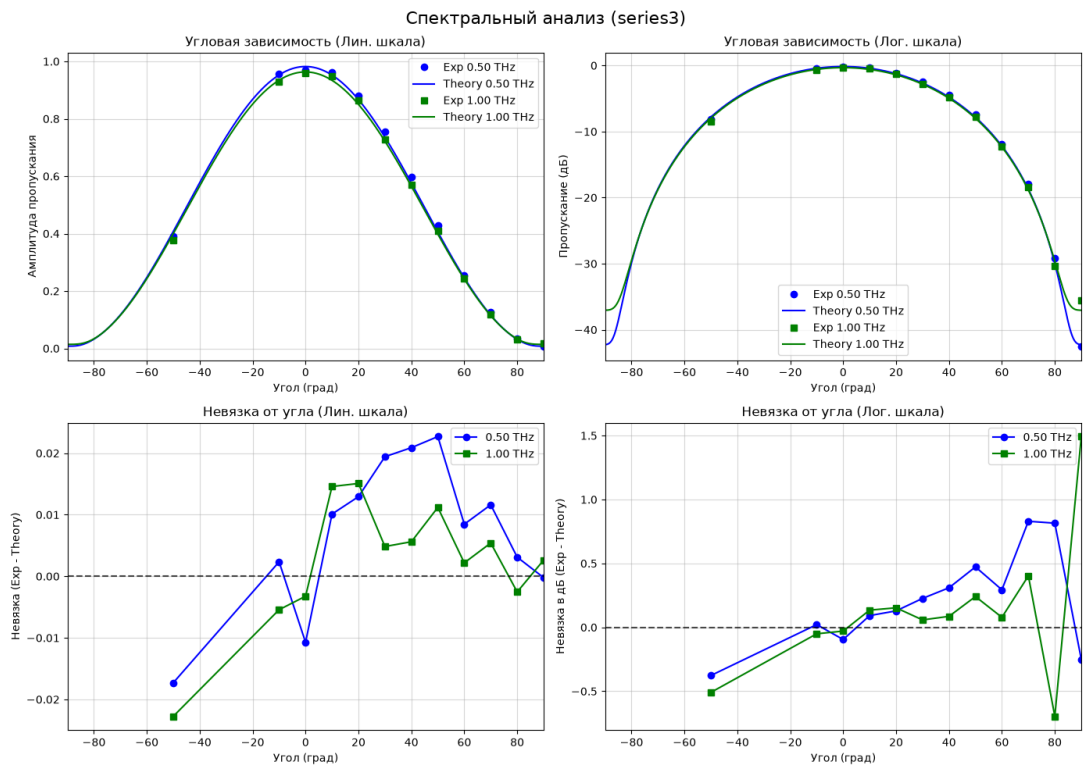
356att.

2.2

series3

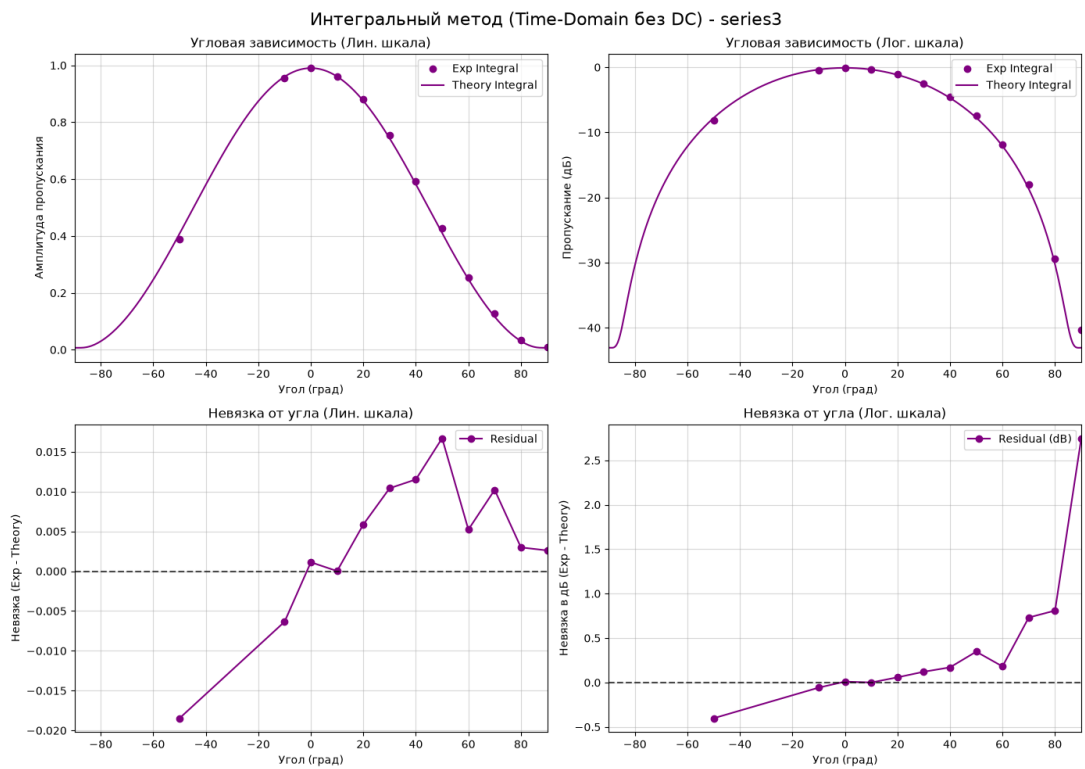
3 4

series3.



. 3:

series3 (0.5 1.0).



. 4:

series3.

3 ()

(ΔA ,) ($\Delta\phi$,)

3.1 (RMSE)

- 356att (2585): $\text{RMSE}_A = 0.950$, $\text{RMSE}_\phi = 0.133$.
- series3 (1384): $\text{RMSE}_A = 0.390$, $\text{RMSE}_\phi = 0.083$.

3.2 (p-value)

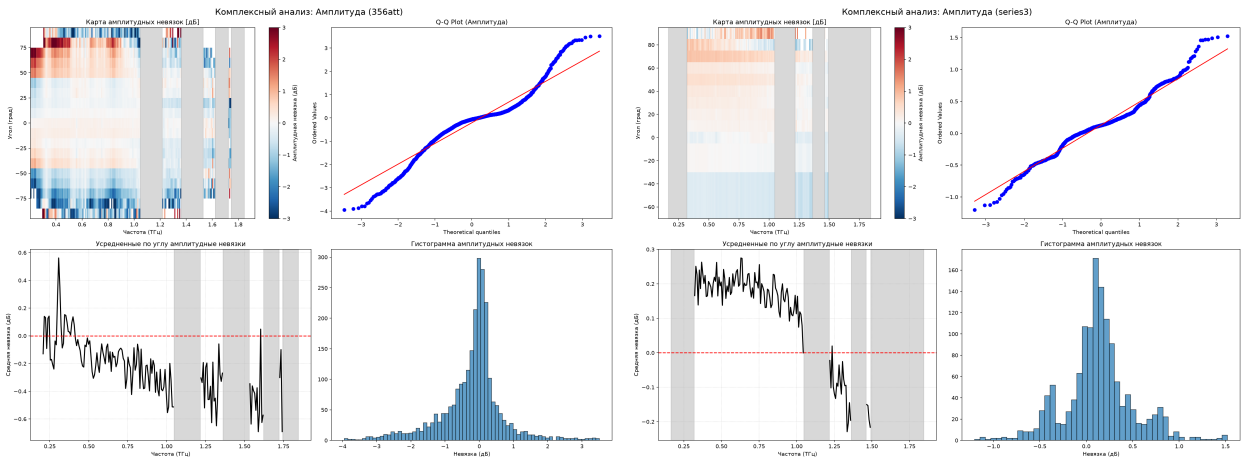
1:

/	- (p -value)	- (p -value)
356att ()	8.42×10^{-36}	4.51×10^{-212}
356att ()	2.35×10^{-39}	9.13×10^{-261}
series3 ()	8.44×10^{-17}	6.15×10^{-27}
series3 ()	1.75×10^{-52}	0.00

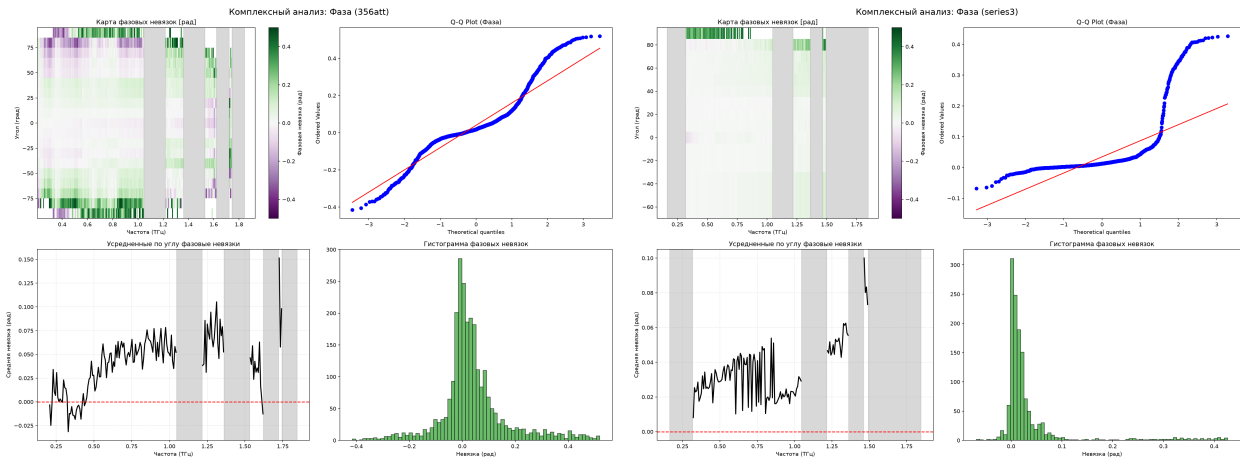
p -value (« »),

3.3 2D

5 6.



. 5: 2D 356att () series3 ().



. 6: 2D 356att () series3 ().

4

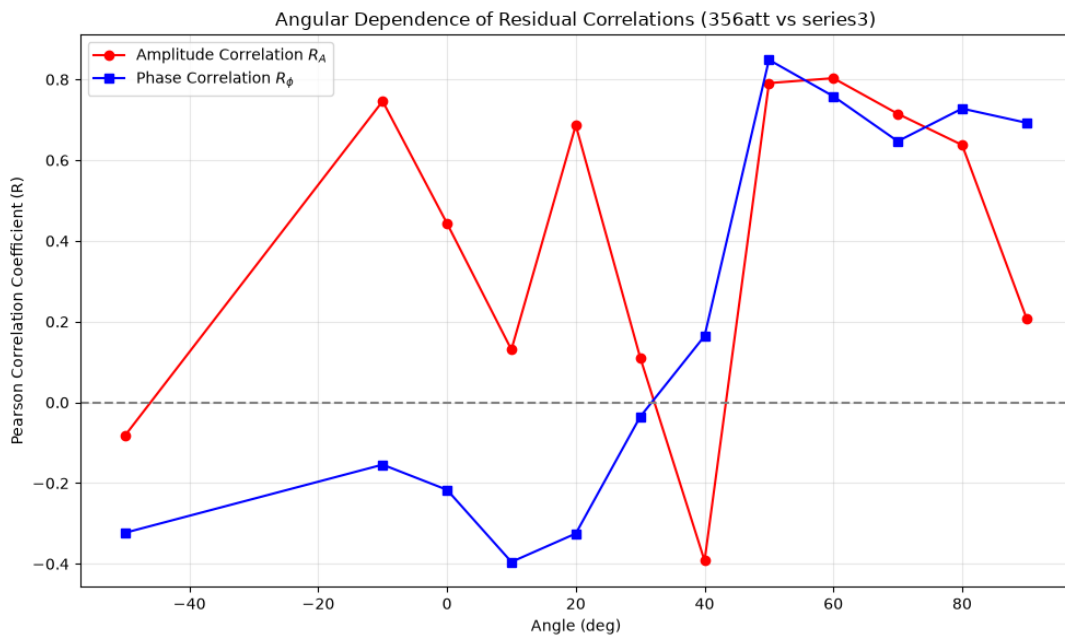
12 (1357 -). 356att series3

4.1

- : Pearson $R = 0.282$ ($p\text{-value} = 2.9 \times 10^{-26}$).
 - : Pearson $R = 0.594$ ($p\text{-value} = 3.1 \times 10^{-130}$).
- (60%) .

4.2 θ

7 .



. 7:

θ .

« » (50° – 90°) (80-85%),
:

- .
- .

—

5

5.1 ($p \approx 0?$)

p -value

- -
Q-Q S-

().

- : ()
- : ,

5.2 (2D-)

2D-

:

1. ($\theta \approx 90^\circ$): - . -
2. ($\nu > 1.5$): - .

5.3 -

356att series3. $(R_\phi \approx 0.594)$ $(R_A \approx 0.282)$
60%, , ,
,

5.4

$(\theta \approx 50^\circ \dots 90^\circ)$ 80–85%[,] :

1. : .
- .
,
2. : P D .
- ,
3. : .
 $(\nu > 1.2 \quad)$,
,

—

6

τ_{ps} - ,
(1).
(FDTD) -